

1. SYLLOGISM (Logical Reasoning)

— Use Venn Diagram

Statements

All locks are keys

All keys are bats



conclusions:-

I. Some bats are locks (T)

II. Some locks are keys (T)

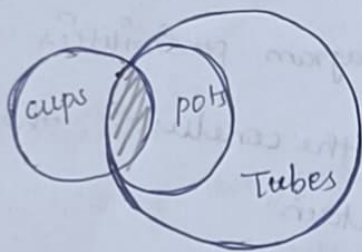
All locks are keys so some locks are keys would also be true.

$\Rightarrow$ Both the conclusions are true.

Statements

Some cups are pots

All pots are tubes



conclusion

I. Some pots are cups (T)

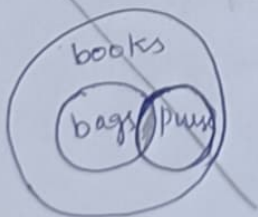
II. Some tubes are cups (T)

$\Rightarrow$ Both the conclusions are true.

Statements

All bags are books

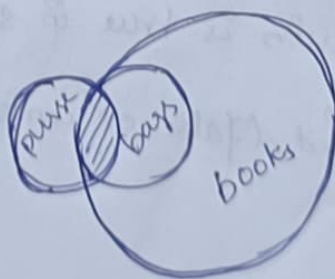
Some purses are bags.



conclusion

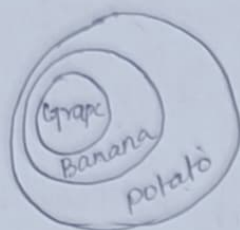
I. Some books are purses (T)

II. Some books are bags (T)



Statements

- All grapes are bananas
- All bananas are potatoes



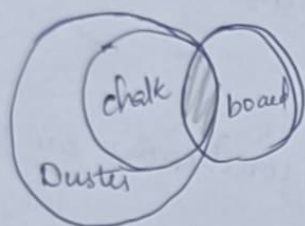
conclusions

- I. Some Potatoes are Bananas T
- II Some Grapes are potatoes T

$\Rightarrow$ Both the conclusions are True

Statements

- All chalks are Dusters
- Some chalks are Boards



conclusion

- I. Some Dusts are Boards T
- II. Some chalks are Dusters T

$\Rightarrow$ Both the conclusions are True

In syllogism if multiple Venn Diagram possibilities exist and even one of them contradicts the conclusion, then the conclusion is considered false or uncertain

Rule:

- $\rightarrow$ If a conclusion must be true in every possible case, it is valid (True)
- $\rightarrow$ If a conclusion is true in some cases but false in others, it is invalid (false or uncertain)

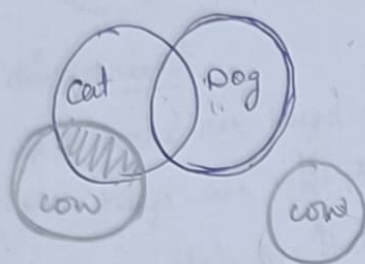
Statements with "No"

Statements

Some cats are dogs

No dog is cow

- 2 possibilities



Conclusion

I. No cow is cat (F)

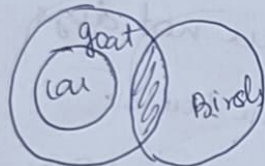
II. No dog is cat (F)

$\Rightarrow$ Both the conclusions are false

Statements

Some goats are Birds

All cars are goats



Conclusions

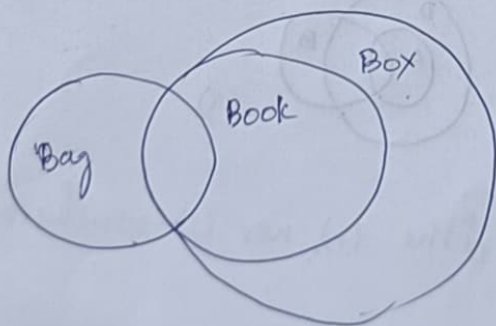
I. Some cars are birds (F)

II. No Bird is Goat (F)

Statements

Some bags are books

All books are bones



Conclusions

I. All bags are bones X (uncertain)

II. No Book is bones X (false)

All trees are flowering trees
Some trees are dogwood!

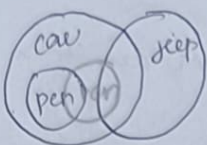
All dogwood are flowering trees

True

Statements

Some cars are jeeps

All pens are cars



Conclusions

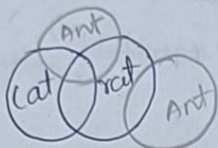
I. No pen is jeep X

II. Some jeeps are cars ✓

Statements

Some cats are rats

Some rats are ants



Conclusions

I. No Rat is Ant X

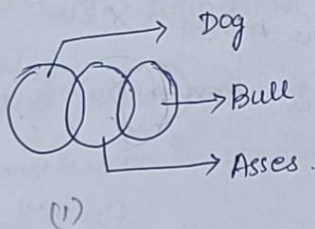
II. No cat is Ant X

cheat: Some-some comes true (maybe)
No cannot be true

Statements

Some dogs are asses

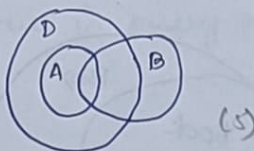
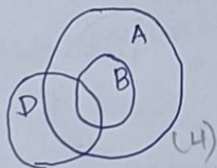
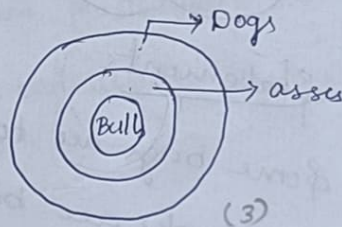
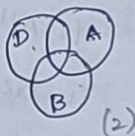
Some asses are bulls



Conclusions

Some asses are not dogs X

Some dogs are bulls



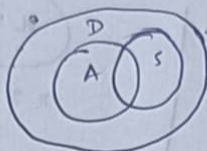
From the dia^s neither (1) nor (2) conclusion follow

Statement

Some actor are singer
All the singer are dancer

Conclusion

- I. Some actor are dancer
- II. No singer is actor



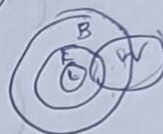
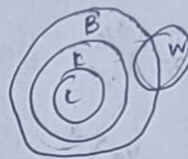
=> only I follows

Statement

All locks are keys.
All keys are bats
Some watch are bats.

Conclusion

1. Some bats are lock
2. Some watch are key
3. All keys are locks

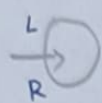


only (1) follows.

PART-1

2. SEATING ARRANGEMENT

people face inside/outside



R & L will
change + W/O

① Count the no of ppl

② Draw frame

③ Sentence-by-sentence divide

④ merge

(1) A, B, C, D, E, F and G are 7 friends sitting in a single row facing North.

1. D is to the immediate right of C

2. E and A are neighbours of F

3. B is to the immediate left of C and on ² second place from left most end.

4. A is at the right most end.

a) Who is between C and E?

D

b) Who are the neighbours of G?

B

c) What is the position of E?

5

5th from left
3rd from right

① C D

② ECA or AFE

③ B C

G	B	C	D	E	F	A
1	2	3	4	5	6	7

left

right



(2) Consider following scenario:

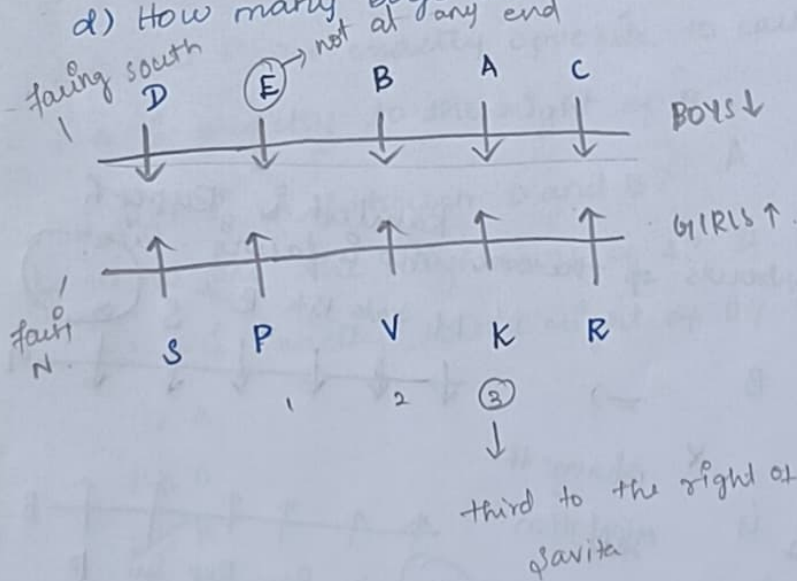
- 5 boys Ashwin, Dipesh, Eshan, Chetan and Bipin and 5 girls Parul, Komal, Radha, Savita and Vimala sit in 2 rows facing towards each other. All the boys are in one row and all the girls in the other row.
- Eshan who is to the immediate right of Bipin and opposite to Parul is not at any end.
- Radha, who is immediate to the right of Komal and opposite to Chetan, is at one of the ends.
- Ashwin is opposite to Komal who is the third to the right of Savita.
- Dipesh and Vimala are not opposite each other.

a) Who is in the middle of the row of boys? Bipin

b) Who is immediately to right of Parul? Vimala

c) Who is opposite of Dipesh? Savita

d) How many boys are there between Ashwin and Bipin? 0



① $\begin{bmatrix} E & B \\ P \end{bmatrix}$ not at any end

② $\begin{bmatrix} C & C \\ K & R \end{bmatrix}$ one of the end
can only be in right end

③ $\begin{bmatrix} A \\ K \end{bmatrix}$

(3) 6 boys Ramesh, Suresh, Dipesh, Rupesh, Paresh, Kamlesh and 6 girls Seema, Geeta, Rita, Meena, Komal, Radha are standing in rows, such that each girl faces one boy, not necessarily in the same order. Seema is to the immediate right of the girl who is facing Paresh, the boy at extreme right. Only Suresh is between Rupesh and Paresh. Kamlesh is to the immediate left of Ramesh and to the immediate right of Dipesh. Rita is facing Ramesh and is to the immediate left of Geeta. Radha is third to the left of Meena.

a) Which girl is facing Rupesh? Radha

b) Which pair of a girl and a boy is at one of the extreme ends?

(i) ~~Komal - Dipesh~~ not necessarily in same end

(ii) Meena - Kamlesh

(iii) Seema - Suresh

(iv) Geeta - Paresh.

c) Who is standing to the immediate left of Rupesh? Ramesh

e) Which option is not same as other 3 options?

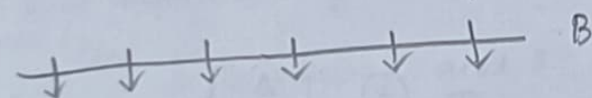
(i) Ramesh - Radha

(ii) Suresh - Komal

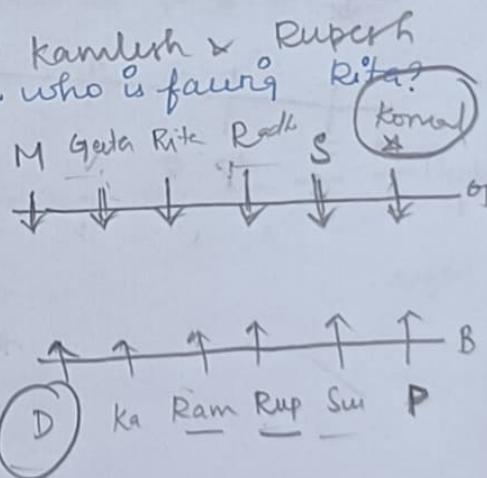
(iii) Dipesh - Geeta

(iv) ~~Rupesh - Radha.~~

f) Who are immediate neighbours of the one who is facing Rita?



X change the orientation



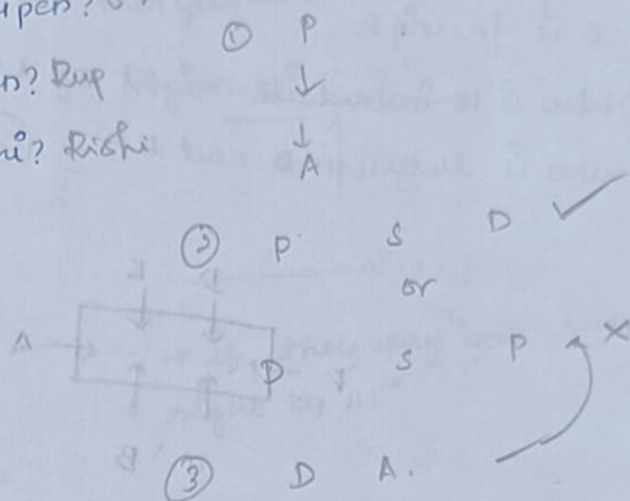
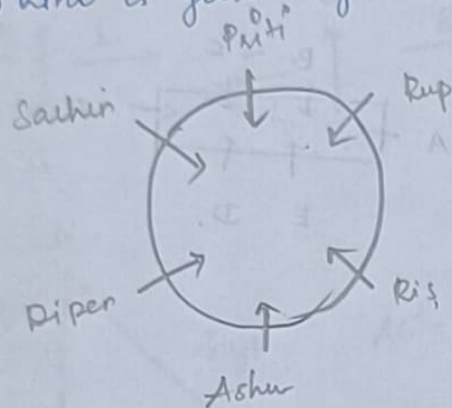
(4) Six friends - Priti, Rupesh, Ashwini, Dipen, Rishi and Sachin, are sitting in a circle facing inwards.

1. Priti and Ashwini are exactly opposite to each other.
2. Sachin is in between Priti and Dipen.
3. Dipen is immediately to the left of Ashwini.
4. Rishi is not exactly opposite to Dipen.

a) Who are the neighbours of Dipen? S & A

b) Who is sitting opposite to Dipen? Rup

c) Who is just right to Ashwini? Rishi



(5) Six friends A, B, C, D, E and F are sitting on a circular table facing outwards.

1. E is between B and C.
2. B and F are exactly opposite to each other.
3. A is exactly to the right of B.

a) Who is in between D and B? A

b) Who are the neighbours of A? D B

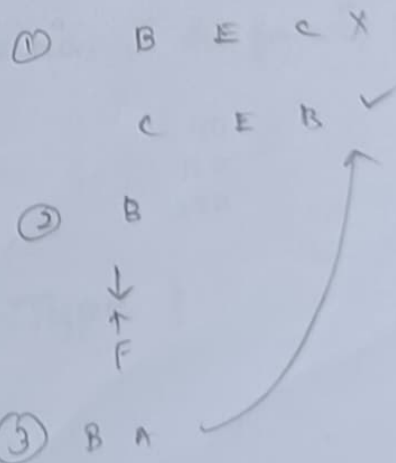
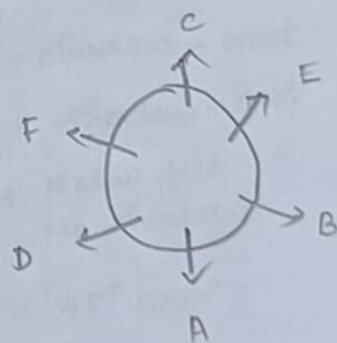
c) Who is exactly to the right of D?

(i) A

(ii) C

(iii) F

(iv) E



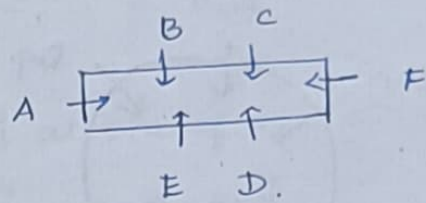
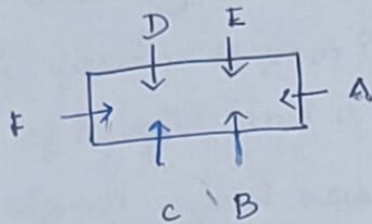
(6) In a group discussion A, B, C, D, E and F are sitting around a rectangular table.

1. A is sitting on a side alone.
2. B, C are sitting on the same side.
3. F is facing A.
4. D is the immediate left of F.
5. E is facing D.

not mentioned inward or outward.

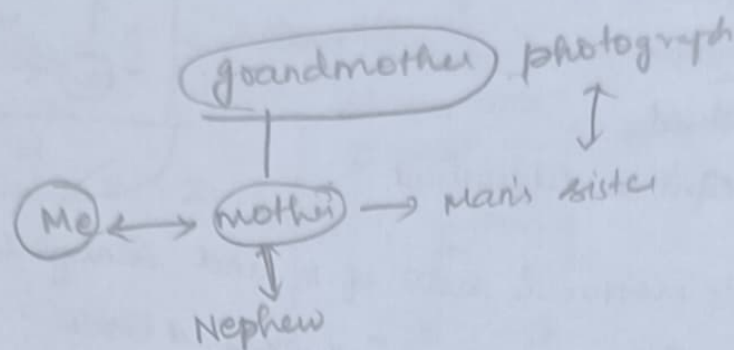
a) Who is to immediate right of A? (E)

b) Who is between B and F? (C)



(U) Pointing to a photograph the man said, "The lady in the photograph is my nephew's maternal grandmother". How is the lady in the photograph related to the man's sister who has no other sister?

-I twist



- (i) cousin
- (ii) sister-in-law
- (iii) Mother
- (iv) Mother-in-law

PART-2 2. SEATING ARRANGEMENT

(T) L, M, N, O, P, Q, R and S are sitting around a square table, such that 4 of them sit at 4 corners of the square, facing centre, while 4 sit in the middle of each of the 4 sides facing outside. 2 females sit in the middle of the sides and 2 at the corners.

1. L sits second to the left of R.

2. R sits in the middle of one of the sides.

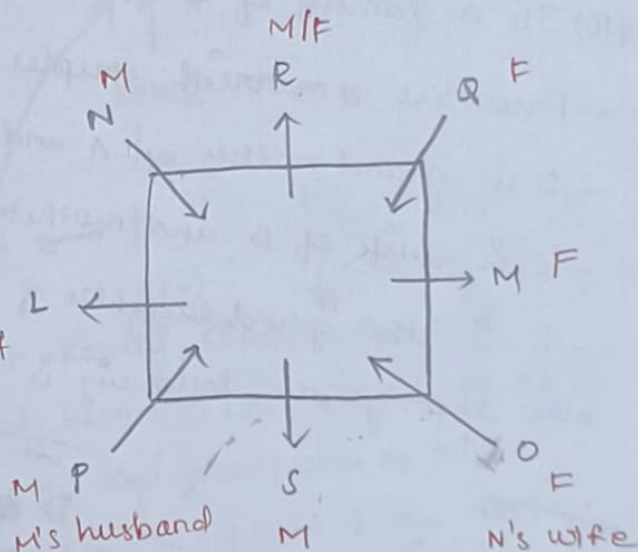
3. N sits fourth to the right of his wife and his wife is not an immediate neighbour of L or R.

4. M sits third to right of her husband

5. M does not sit at any of the corners

6. Only O sits between M & S.

7. S is husband of L and P is a male.



- check all possibilities
- use male & female hint
- every line & word is a clue read carefully.

a) Which of the following is true with respect to men seating arrangement?

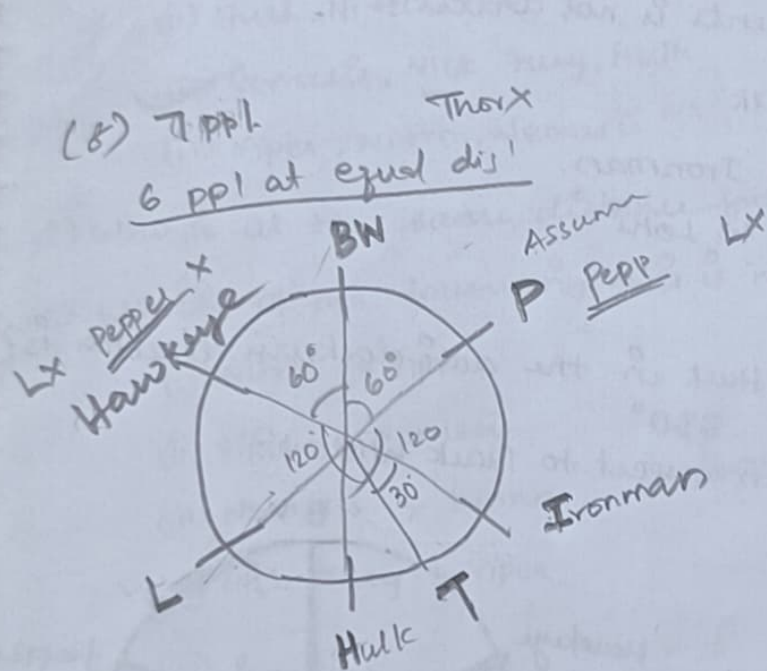
- (i) No two males are immediate neighbours of each other
- (ii) R and S face each other in the seating arrangement.
- (iii) L sits in the centre of one of the sides of the square table
- (iv) Q is a male and sits diagonally opposite to P.

b) Who is M's husband? P

c) How many people sit between M and N when counted in anti-clockwise direction from M? 2

d) Who is the wife of N? O

e) What is the position of P with respect to N? 2nd pos to the right of N.



By default if nothing is given everyone will face inside

(8) Read the following information

1. Pepper Potts, Hulk, Ironman, Hawkeye, Thor, Loki and Black Widow are sitting in a circle.
2. Pepper, Hulk, Ironman, Hawkeye, Loki and Black Widow are sitting at equal distances from each other.
3. Ironman is sitting two places right of Loki, who is sitting one place right of Hawkeye.
4. Pepper forms an angle of 90° from Thor and an angle of 120° from Hulk.
5. Hulk is just opposite to Black Widow and is sitting on the left of Thor.

(a) Who is the only person sitting on the immediate left of Black Widow?
PP

(b) Which of the following statements is not correct?

- (i) Loki is on the right of Hulk
- (ii) Hawkeye is sitting next to Ironman.
- (iii) Thor is sitting adjacent to Loki
- ~~(iv) All of the above.~~

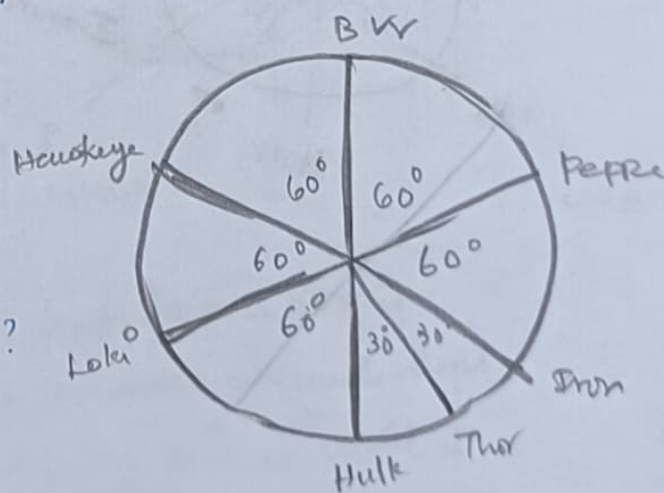
(c) The angle between Thor and Hulk in the anti-clockwise direction is?
 330°

(d) Where is Hawkeye sitting with respect to Black Widow?

- ~~(i) To immediate right~~
- (ii) To immediate left
- (iii) Exactly opposite
- (iv) None of these

(e) Thor is sitting between which pair?

Hulk & I.M



9) Read the following.

1. 6 people Nitro, Nemesis, Nick Fury, Viper, Kronos and Hulk are attending a hexagonal table conference.

2. All the sides of the hexagon table so formed are of same length.

3. Nitro is not adjacent to Nemesis or Nick Fury, Viper is not adjacent to Nick Fury or Kronos.

4. Nemesis and Nick Fury are adjacent, Hulk is in the middle of Viper and Nick Fury.

a) Who is placed exactly opposite to Kronos? HULK

b) If one neighbor of Nitro is Viper, who is the other one? KRONOS

c) Which of the following are in the right sequence?

(i) Nitro, Hulk, Nemesis

(ii) Hulk, Nitro, Kronos

☒ (iii) Nemesis, Nick Fury, Hulk

(iv) Viper, Nitro, Nemesis

d) Who is at the same distance from Viper as Kronos is from Viper? N.F

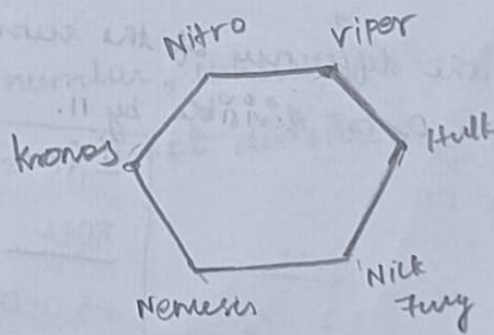
e) Which of the following pairs is neighbour pair of Hulk?

(i) Nitro & Viper

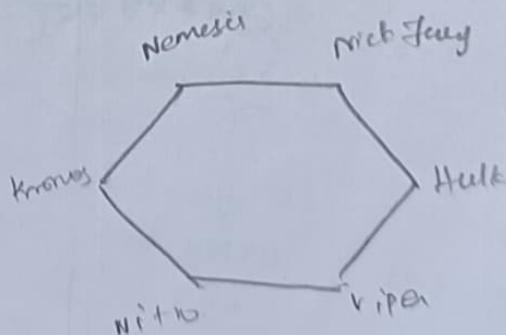
(ii) Viper & Nemesis

(iii) Nemesis & Kronos

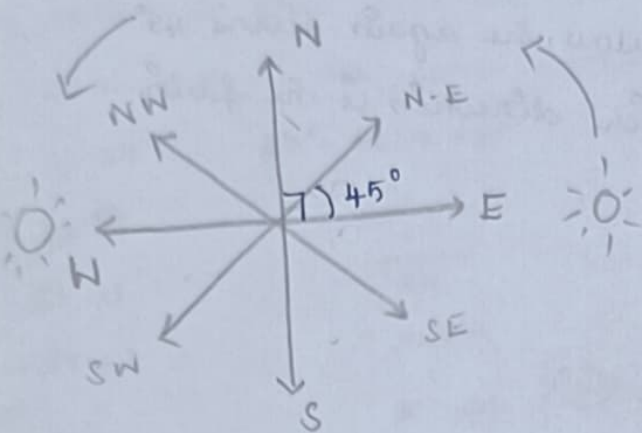
☒ (iv) Nick Fury & Viper



You can start from any arbitrary value



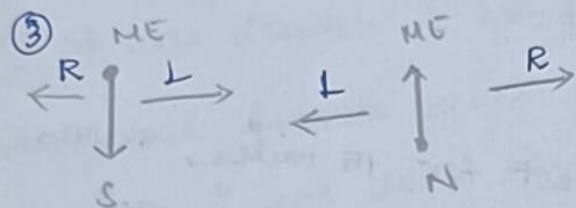
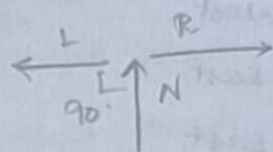
3. DIRECTION TEST



ANGLES

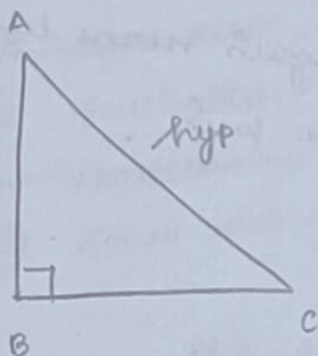
① When angles are not mentioned by default it is 90° .

→ you travel towards north and then you take a left or right



→ if they say you turn right by 45°

④



$$AC^2 = AB^2 + BC^2$$

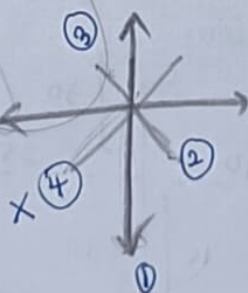
② SHADOW



Morning → shadow - west
Evening → shadow - east
12 noon → Below feet
No shadow

① Joe is facing towards South and turns 45° anti-clockwise. He turns again 180° in anti-clockwise direction. Now, he turns 270° clockwise. Which direction is he facing?

- (a) West
- (b) East
- (c) South-West
- (d) North-East

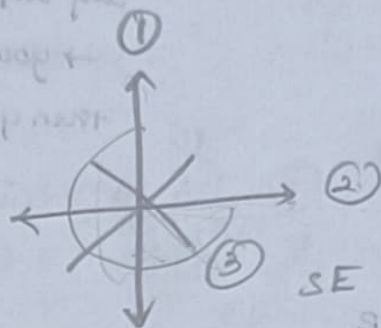


90
180
270

Type 1

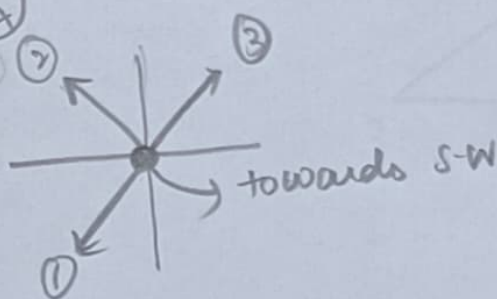
② Morris is facing North and walks 10 kms. He turns 270° anti-clockwise and walks 15 kms. Now, he again turns 45° clockwise and walks for 25 kms. Which direction is he facing now?

- a) North-West
- b) South-West
- c) North-East
- d) South-East
- e) None of these



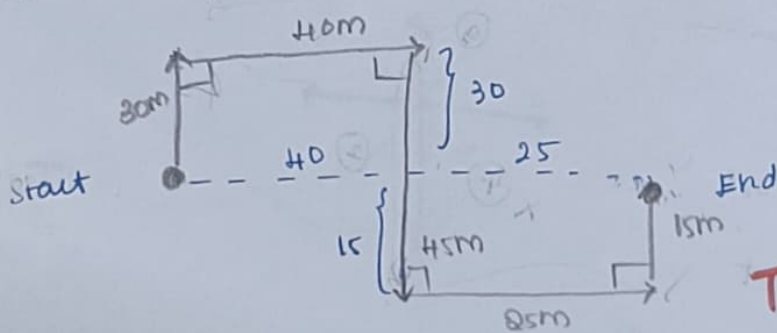
③ Mary is walking towards South-West for 15 miles, turns right and walks another 10 miles. She then turns 90° clockwise and walks 10 miles. Now, she again turns left and walks 30 miles which direction is she facing?

- a) East
- b) West
- c) South East
- d) North-West
- e) can't determine



④ Maria walked 30m towards north. She turned right and walked 40m. She then turned right and walked 45m. She turned left and walked 25m. Finally she turned left and walked 15m. How far is she from the starting position?

- a) 55
- b) 65
- c) 30
- d) 45



$$40 + 25 = 65$$

Type II

⑤ A Boy walks 12 kms towards east. He turns 90° clockwise and walks 20 kms then he turned right and walked for 33 kms. How far is he from starting point?

a) 65

b) 29

c) 21

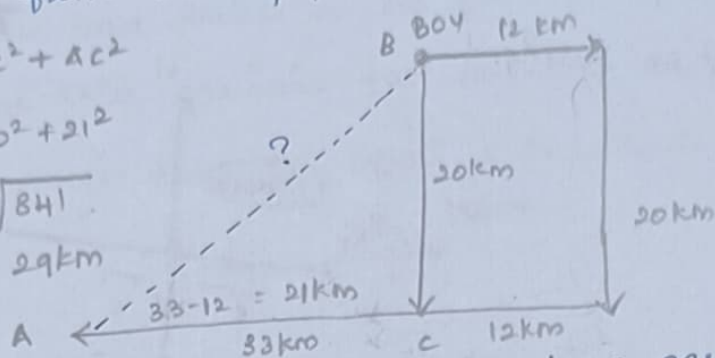
d) 43

$$AB^2 = BC^2 + AC^2$$

$$= 20^2 + 21^2$$

$$= \sqrt{841}$$

$$= 29 \text{ km}$$



⑥ A car travels 25 kms towards south from garage. It turns left and travels 30 kms, then turns right and travels 15 kms. How far is car from the garage and in which direction?

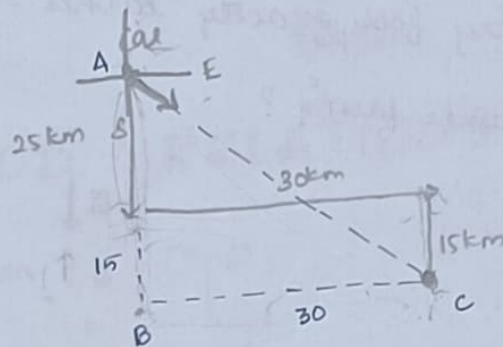
Type III

a) 40 North-East

b) 70 South-East

c) 60 North-East

d) 50 South-East



$$AC^2 = AB^2 + BC^2$$

$$= 40^2 + 30^2$$

$$= 2500$$

$$AC = 50$$

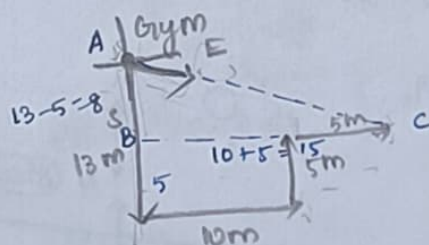
⑦ Martin walked 13 meters towards south from his Gym. He turned left and walked 10 meters. He took a left turn and walked 5 meters and then he turned right and walked for 5 m. How far is he from his Gym and in which direction?

a) 19 North-West

b) 13 South-East

c) 15 North-West

d) 17 South-East



$$AC^2 = AB^2 + BC^2$$

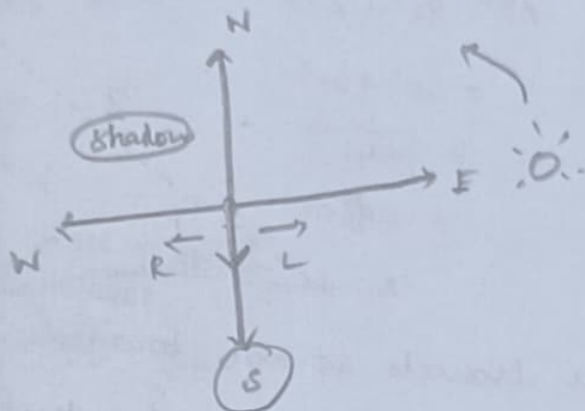
$$= 64 + 225$$

$$= 289 \quad 17 \text{ m}$$

⑧ One morning after sunrise, Joe was standing facing a pole. The shadow of the pole fell exactly to his right. To which direction was he facing?

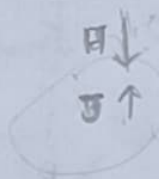
Type - iv SHADOWS

- a) East
- b) ~~South~~
- c) West
- d) Data is inadequate



⑨ James and Henry were standing facing each other at 8 am. Shadow of the Henry fell exactly to his right. To which direction was James facing?

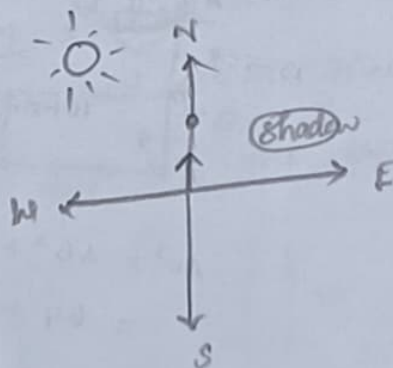
- a) East
- b) South
- c) ~~North~~
- d) Data is inadequate



Henry is facing South
So James is facing North

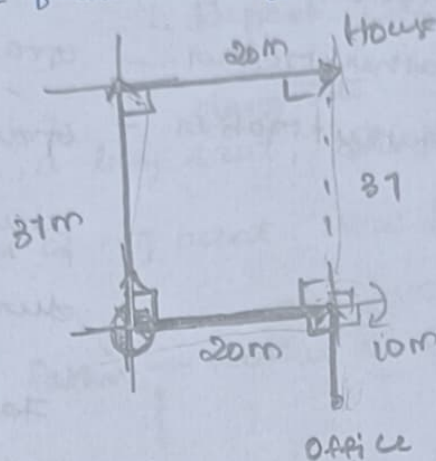
⑩ Prince was standing facing to the pole at 1:48 pm. Shadow of the pole fell towards his right. To which direction was Prince facing?

- a) West
- b) South
- c) East
- d) ~~North~~



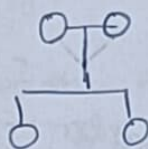
11) Ted drives a car 10 miles towards North from his office. He turns 270° clockwise and drives for 20 miles. Now, he turns 270° anti-clockwise and drives a car for 31 miles. Finally, he reaches his house after driving for 20 miles to his right. **!!!**
How far is Ted's house from his office & in which direction?

- a) 40 South
- b) 41 North
- c) 30 East
- d) 31 West.



4. BLOOD RELATIONS

① Draw the diagram



② New level for another generation

③ Use simple symbols. (Eg $+$ $\rightarrow$ M $-$ $\rightarrow$ F)

④ Don't judge gender based on name

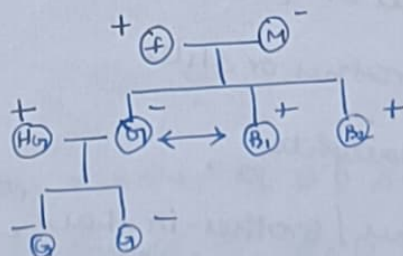
⑤ Symbols

$+$ $\rightarrow$ male

$-$ $\rightarrow$ female

$\bigcirc - \bigcirc \rightarrow$ married

$\bigcirc \leftrightarrow \bigcirc \rightarrow$ siblings.



Relations:-

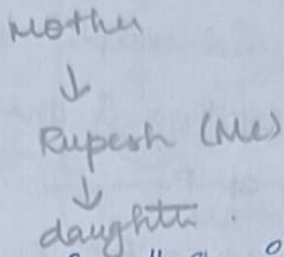
Father of Father or Mother	- Grand Father
Mother of Father or Mother	- Grand Mother
Wife of Grand Father	- Grand Mother
Husband of Grand Mother	- Grand Father
Father-in-law of Father/Mother	- Grand Father
Mother-in-law of Father/Mother	- Grand Mother
Brother of Father	- Uncle
Sister of Father	- Aunt

Father side → Paternal
Mother side → Maternal

Brother / Sister of Mother	- Maternal Uncle / Aunt
Father of Wife / Husband	- Father-in-law
Mother of Wife / Husband	- Mother-in-law
Father's / Mother's only son / daughter	- oneself & You
Son / Daughter of Uncle / Aunt	- Cousin
Brother / Sister of Husband / Wife	- Brother-in-law / Sister-in-law
Husband of Sister / Sister-in-law	- Brother-in-law
Son of Brother or Sister	- Nephew
Daughter of Brother or Sister	- Niece
Husband of Daughter	- Son-in-law
Wife of Brother / Brother-in-law	- Sister-in-law
Wife of Son	- Daughter-in-law

① Rupesh introduces Poonam as the [daughter of the only son of my father's wife]. How is Poonam related to Rupesh?

- i) cousin
- ii) Niece
- iii) Daughter
- iv) Aunt



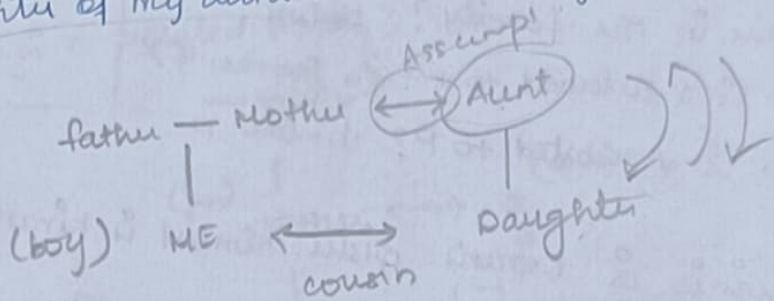
② Introducing a girl, a boy said, "She is the daughter of the mother of the daughter of my aunt." How is the girl related to the boy?

father — mother — Aunt

Assumpt

Diagram illustrating the relationship: A horizontal line connects 'father' and 'mother'. A vertical line descends from 'mother' to 'Aunt'. A double-headed arrow connects 'mother' and 'Aunt'. To the right of 'Aunt' are two vertical lines, one with a downward arrow and a circle at the bottom, and another with a downward arrow.

- i) ~~Cousin~~
- ii) Niece
- iii) Daughter
- iv) Aunt.



iv) Aunt.

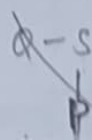
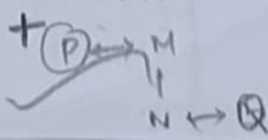
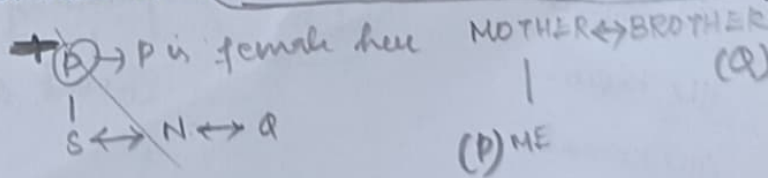
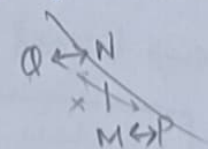
③ $U+V$ means U is the brother of V , $W-x$ means W is the father of S , $x \div Y$ means x is the sister of Y , $Y \times Z$ means Z is the mother of Y . Which of the following means that N is the mother of O .

- Visualize
- (i) $L + M \div N - O$
- (ii) $\underline{L - M \times O \div P}$ N is not here
- (iii) $N \div M \times L \div O$
- (iv) $M + L \div \underline{O \times N}$

④ If $A \# B$ means A is the mother of B ; $A - B$ means A is brother of B ; $A \% B$ means A is father of B ; $A \times B$ means A is sister of B , which of the following shows that P is the maternal uncle of Q ?

~~1) $P \# Q$~~ $\rightarrow P$ is female here MOTHER $\leftrightarrow$ BROTHER

- (i) ~~$Q - N \# M \times P$~~
(ii) ~~$P \# S \times N - Q$~~
(iii) ~~$P - M \# N \times Q$~~
(iv) ~~$Q - S \% P$~~



(5) Read the information carefully & answer the questions.

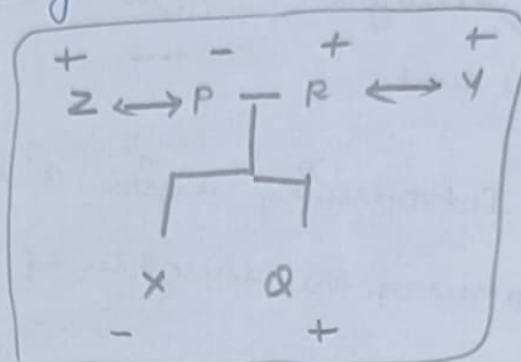
1. A family consists of 6 members P, Q, R, X, Y, Z

2. Q is the son of R but R is not mother of Q iii family tree

3. P and R are married couple

4. Y is the brother of R, X is the daughter of P

5. Z is the brother of P.



Q1) Who is the mother-in-law of R? Z

2) How many female members are there in the family? 2

Q3) How is Q related to X? ~~Brother~~ Sibling,
 Y Brother

Q4) How is Y related to P? in-law P

(6) Ashwini is Kiran's sister. Komal is Kiran's mother. Dipen is Komal's father. Esha is Dipen's mother. Then, how is Ashwini related to Dipen?

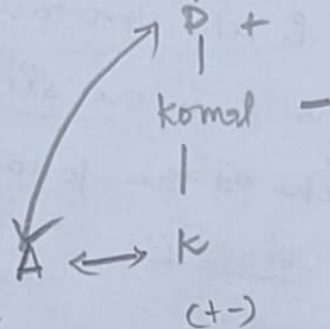
Ashwini is granddaughter of Dipen

(i) Grandfather

(ii) Grandmother

(iii) Daughter

(iv) Granddaughter



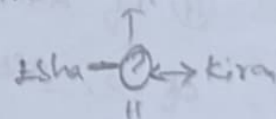
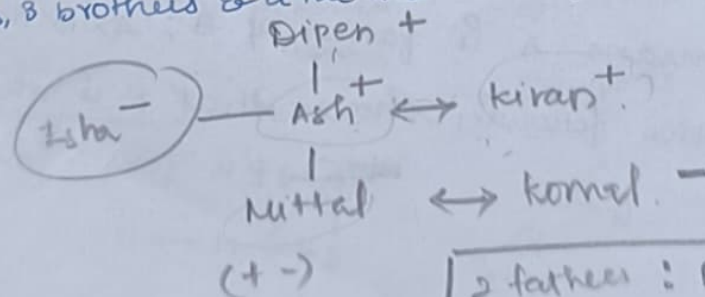
(7) There are 6 persons Ashwini, Kiran, Komal, Dipen, Esha and Mittal. Komal is the sister of Mittal. Kiran is the brother of Esha's husband. Dipen is the father of Ashwini and grandfather of Mittal. There are 2 fathers, 3 brothers & a mother in the grp. Who is the mother?
 This must be Ashwini

(i) Ashwini

(ii) Kiran

(iii) Komal

(iv) Esha



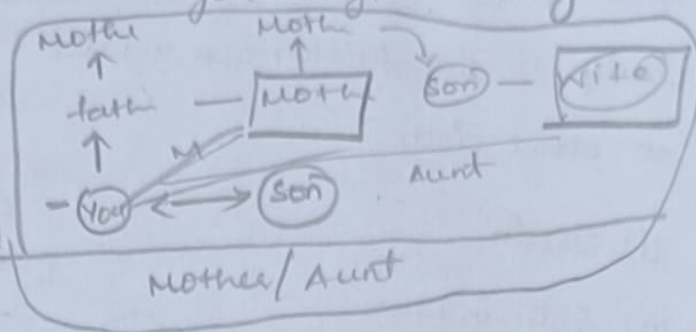
2 fathers : Dipen, Ashwini

3 brothers : Ashwini, Kiran, Mittal

Mother : Esha

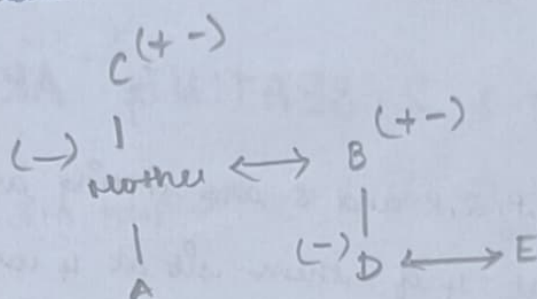
(8) Pointing out to a lady, a girl said, "She is the daughter-in-law of the grandmother of my father's only son." How is the lady related to the girl?

- Sister-in-law
- Mother
- Aunt
- can't be determined.



(9) A's mother is sister of B and daughter of C. D is daughter of B and sister of E. How is C related to E?

- Father
- Grandfather
- Grandmother
- Either grandfather or grandmother



(10) In a family of six persons A, B, C, D, E and F

- There are 2 married couples.

- D is grandmother of A and mother of B.

- C is wife of B and mother of F.

- F is the granddaughter of E.

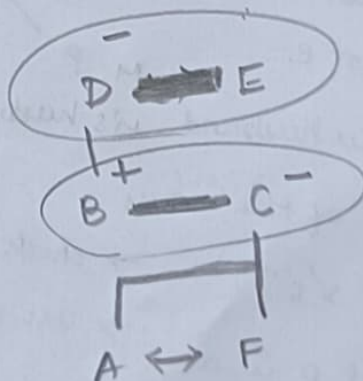
Who among the following is one of the couples?

(i) CD

(ii) DE

(iii) EF

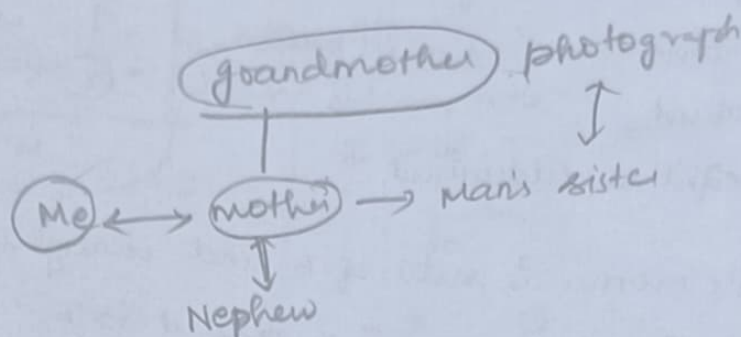
(iv) None of these



(11) Pointing to a photograph the man said, "The lady in the photograph is my nephew's maternal grandmother". How is the lady in the photograph related to the man's sister who has no other sister?

-I twist

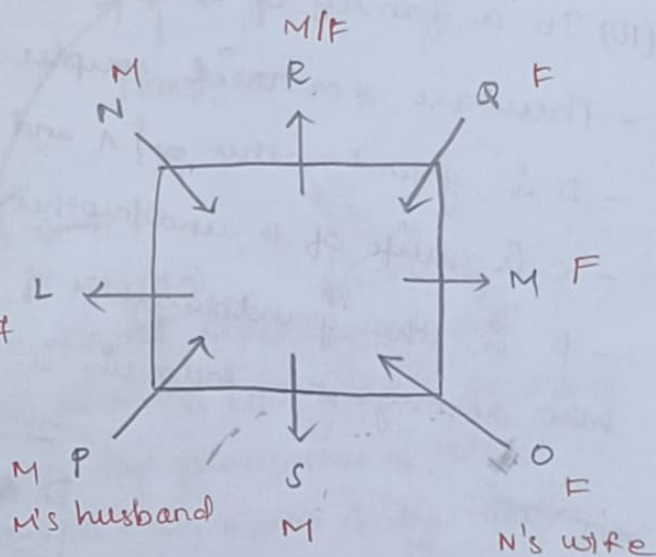
- (i) cousin
- (ii) sister-in-law
- (iii) Mother
- (iv) Mother-in-law



PART-2 2. SEATING ARRANGEMENT

(1) L, M, N, O, P, Q, R and S are sitting around a square table, such that 4 of them sit at 4 corners of the square, facing centre, while 4 sit in the middle of each of the 4 sides facing outside. 2 females sit in the middle of the sides and 2 at the corners.

1. L sits second to the left of R.
2. R sits in the middle of one of the sides.
3. N sits fourth to the right of his wife and his wife is not an immediate neighbour of L or R.



4. M sits third to right of her husband
5. M does not sit at any of the corners
6. Only O sits between M & S.
7. S is husband of L and P is a male.

- check all possibilities
- use male & female hint
- every line & word is a clue read carefully.